

Background

Convergence Insufficiency (CI) is a prevalent binocular vision disorder, reported in the United States with rates ranging from 2.25% to 17.6% [1]. CI is typified by exophoria—an outward eye drift when focusing on near objects—a receded near point of convergence (NPC), and an insufficient positive fusional vergence range (PFV) [2].

These conditions culminate in a significant challenge: patients with CI often *experience difficulties in reading*, frequently losing their place on the page or struggling with double vision. This suggests an essential link between visual capabilities and reading skills, a connection further evidenced by a high prevalence of visual disorders among individuals reported with reading disabilities [3,4].

Our research builds on this observed correlation, exploring the complex interrelationship between CI and reading abilities, with a specific interest in understanding the neurological implications of this relationship.

To this end, we leverage Diffusion Tensor Imaging (DTI) to investigate the white matter structures in a group of CI patients. We pay particular attention to the *arcuate fasciculus*, a crucial neural tract connecting language regions in the frontal cortex with visual regions in the inferior temporal cortex [5]. By studying this neural pathway, we aim to provide a deeper understanding of CI's impact on reading abilities.

Methods

Subjects

6 CI patients and 6 controls, with age range 18~30 years old.
Convergence Insufficiency Symptom Survey (CISS): ≥ 21 for adult CI
Near Point of Convergence (NPC): ≥ 6 cm for CI
Positive Fusional Vergence (PFV) at near: ≤ 15 PD BO for CI
Exophoria at Near - Exophoria at Distance: ≥ 4 PD for CI

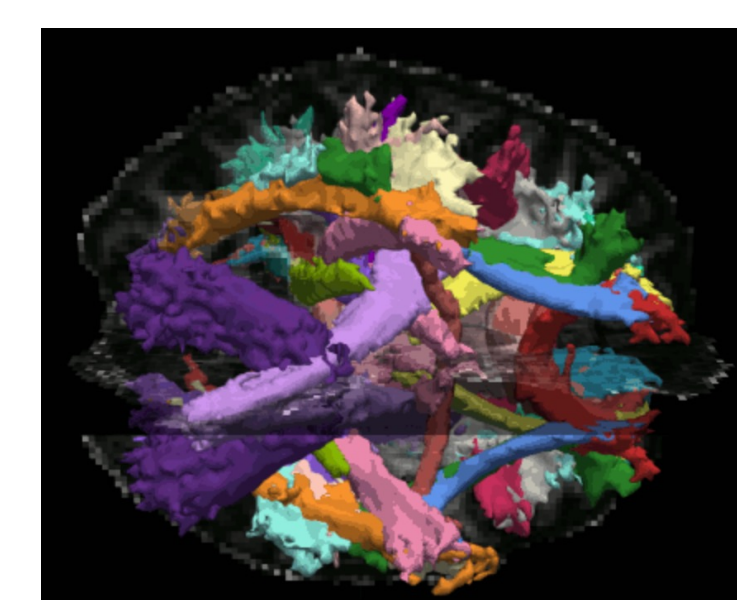
Neuroimaging

Participants were scanned at the Center for Cognitive and Behavioral Brain Imaging (CCBBI). A high-resolution T1-weighted image was acquired (1mm isotropic, TR/TE = 1950/4.44 ms, flip angle = 12°, matrix size = 72 × 72), as well as a 64-direction diffusion-weighted series (2mm isotropic, b=900 s/mm²)

Analysis

TRACULA (TRActs Constrained by UnderLying Anatomy) was used to reconstruct white matter tracts from each participant's native diffusion images. The reconstruction relies on each participant's cortical parcellation, subcortical, and thalamic nuclei segmentation.

This automatic reconstruction has proven to have higher sensitivity and specificity than traditional manual labelling[6].



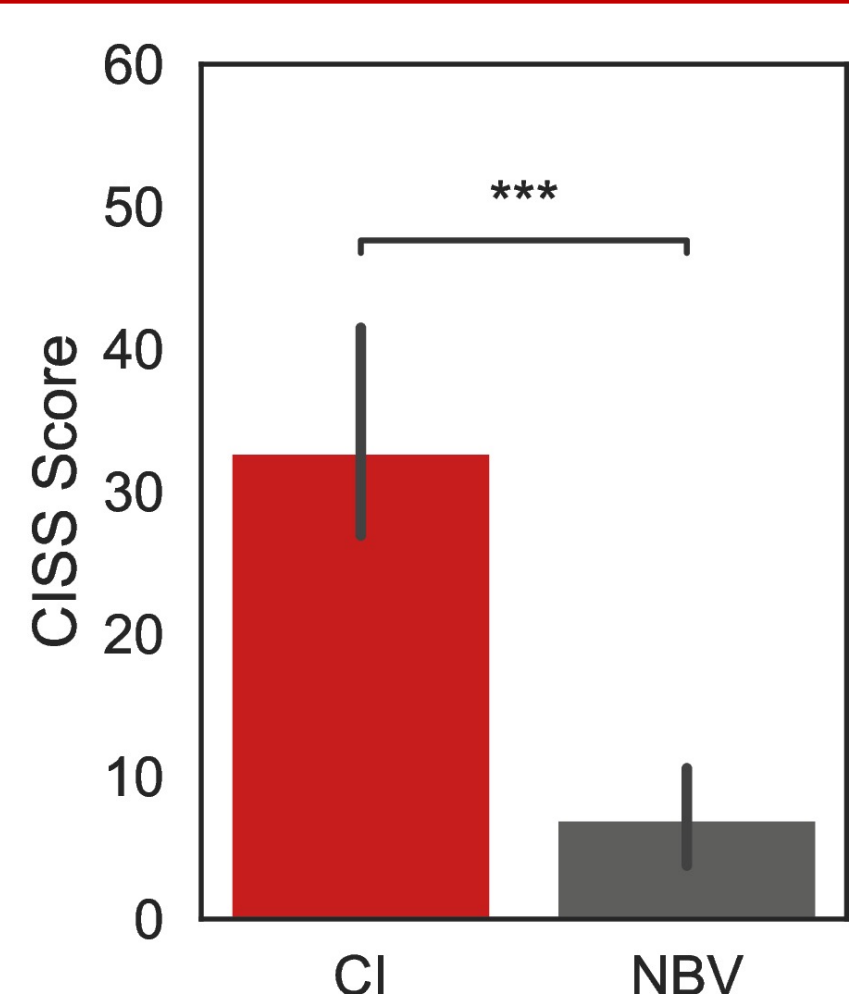
Example of TRACULA reconstruction

Reading Symptoms

Controls vs. Patients in reading ability

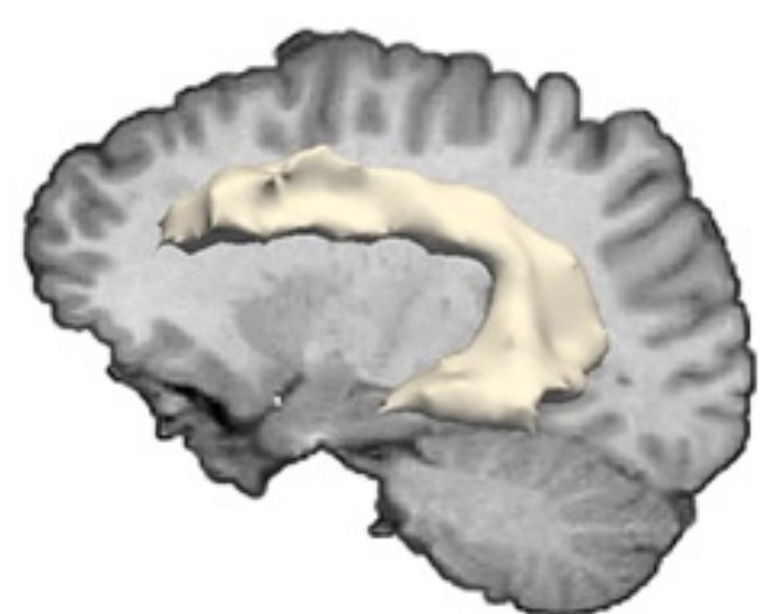
Controls and CI patients show great *difference* in reading-related symptoms.

Controls: 6.83±4.56
Patients: 33.25±9.28
p=1.957e-04



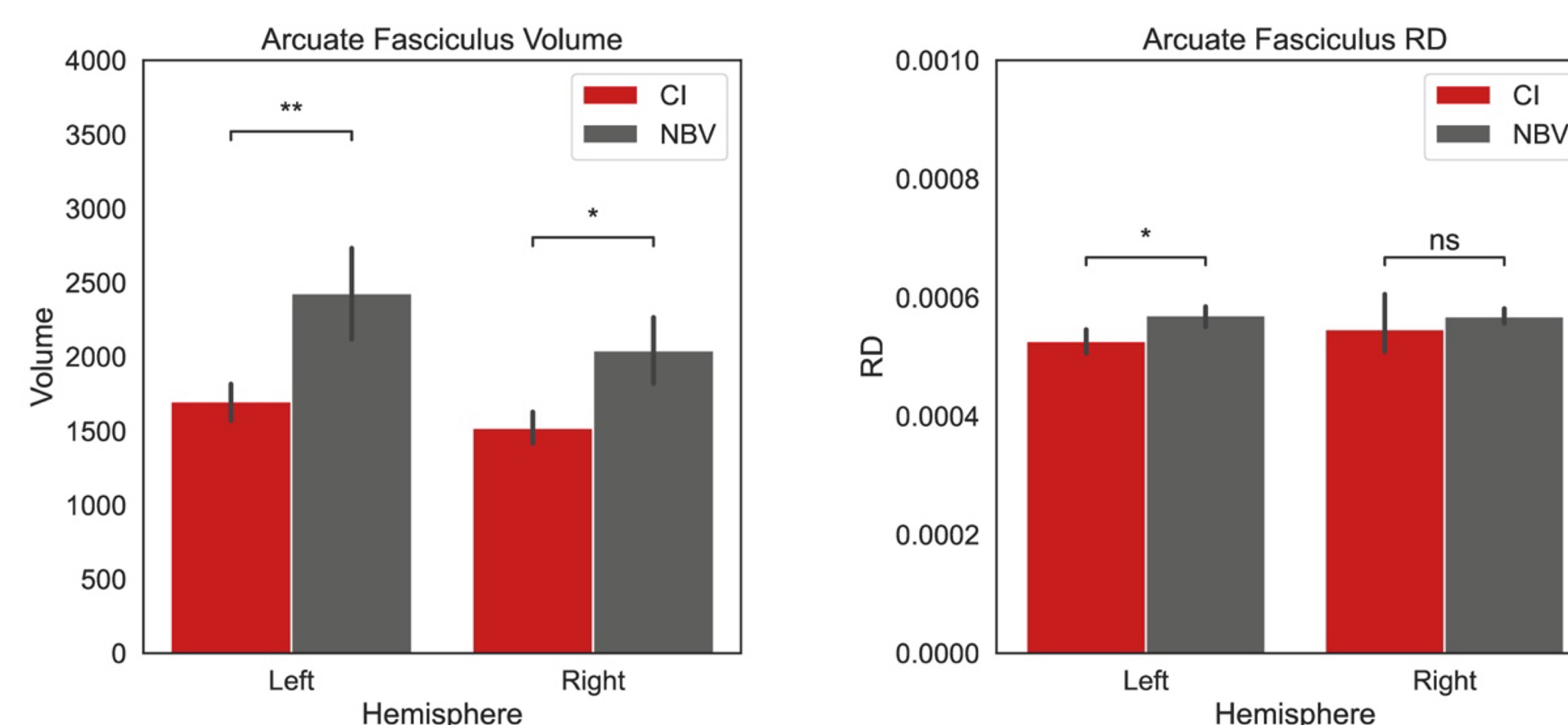
Reading and left arcuate fasciculus

The left arcuate is part of the left superior longitudinal fasciculus and connects temporoparietal and inferior frontal regions that are core constituents of the language network.

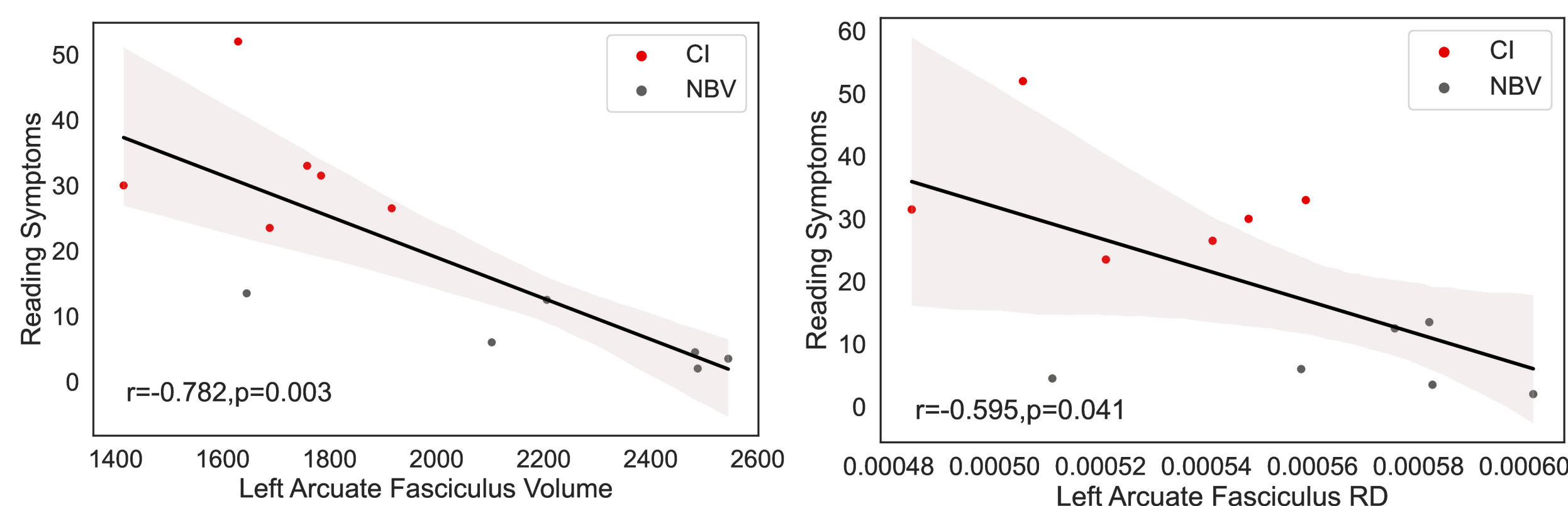


Smaller Arcuate, Lower RD

CI patients have *smaller arcuate fasciculi* in both hemispheres
They also have *lower radial diffusivity (RD)* compared to controls in the left hemisphere

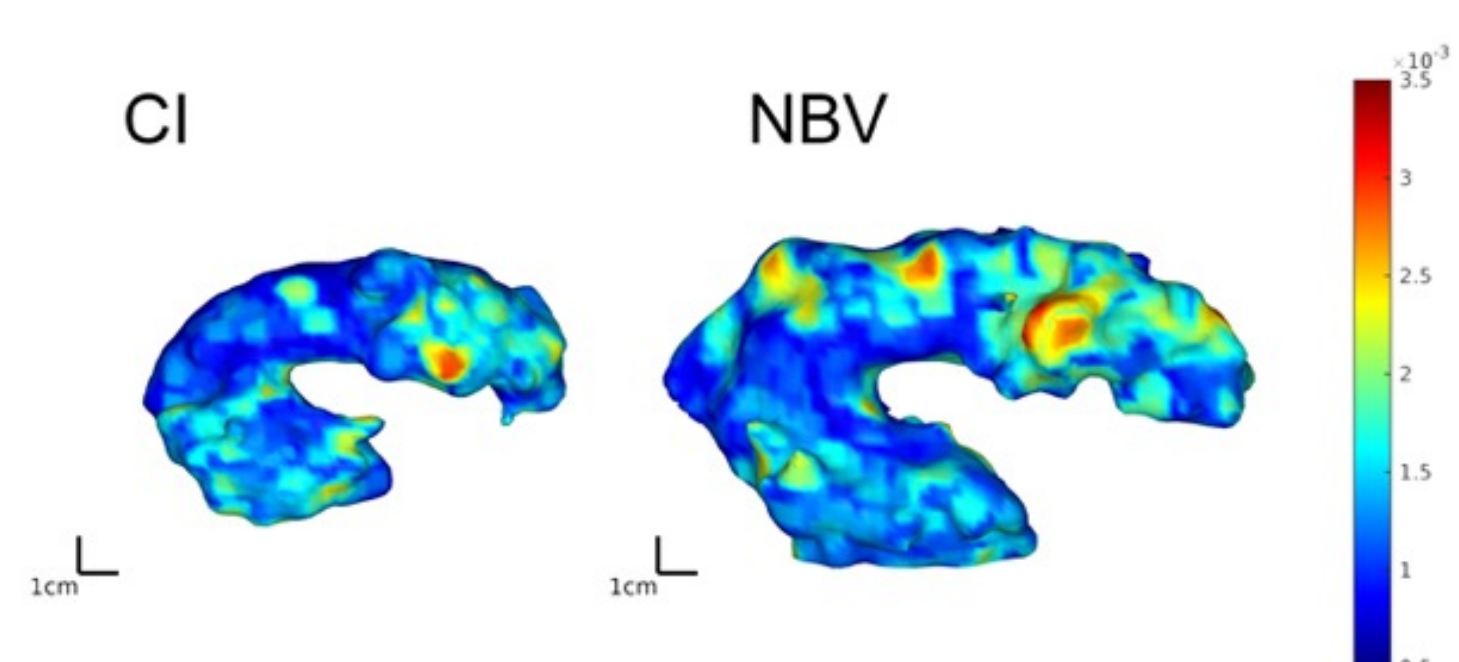


Reading and the Arcuate



Reading-related symptoms are *significantly correlated* with arcuate *volume*, as well as *RD*, which has previously been reported in healthy controls[5,7]. Individuals with lower arcuate volume and RD tend to be poorer readers.

This suggests that CI may lead to abnormal white matter integrity in the arcuate fasciculus, possibly caused by reduced reading experience.



CI patients have smaller arcuate fasciculi with lower RD

Conclusions

- CI patients demonstrate a *higher incidence of reading deficits*, signaling a direct correlation with reading challenges.
- In CI patients, *the volume and RD of the left arcuate fasciculus*, a key language-processing network, are *reduced*, implying CI's impact on these brain structures.
- Our study shows a negative *correlation* between reading *symptom* severity and both the *volume and RD* of the *left arcuate fasciculus*, linking these structural variations to reading abilities.

In conclusion, our findings indicate that **CI may induce reading difficulties by altering associated brain networks.**

References

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